

6.MERGE SORT :

```
#include <stdio.h>
```

```
void merge(int a[], int l, int m, int r) {  
    int i = l, j = m + 1, k = 0, temp[100];
```

```
    while (i <= m && j <= r) {  
        if (a[i] <= a[j])  
            temp[k++] = a[i++];  
        else  
            temp[k++] = a[j++];  
    }
```

```
    while (i <= m)  
        temp[k++] = a[i++];
```

```
    while (j <= r)  
        temp[k++] = a[j++];
```

```
    for (i = l, k = 0; i <= r; i++, k++)  
        a[i] = temp[k];  
}
```

```
void mergeSort(int a[], int l, int r) {  
    if (l < r) {  
        int m = (l + r) / 2;  
  
        mergeSort(a, l, m);  
        mergeSort(a, m + 1, r);
```

```
        merge(a, l, m, r);
    }
}

int main() {
    int n, a[100];

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    mergeSort(a, 0, n - 1);

    printf("Sorted array:\n");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

OUTPUT:

main.c



Share

Run

Output

```
1 #include <stdio.h>
2
3 void merge(int a[], int l, int m, int r) {
4     int i = l, j = m + 1, k = 0, temp[100];
5
6     while (i <= m && j <= r) {
7         if (a[i] <= a[j])
8             temp[k++] = a[i++];
9         else
10            temp[k++] = a[j++];
11    }
12
13    while (i <= m)
14        temp[k++] = a[i++];
15
16    while (j <= r)
17        temp[k++] = a[j++];
18
19    for (i = l, k = 0; i <= r; i++, k++)
20        a[i] = temp[k];
21 }
22
```

```
Enter number of elements: 5
Enter elements:
1 45 2 26 3
Sorted array:
1 2 3 26 45
```

```
=== Code Execution Successful ===
```